Benchmarking Foundation Models using Satellite Image Time Series for land monitoring

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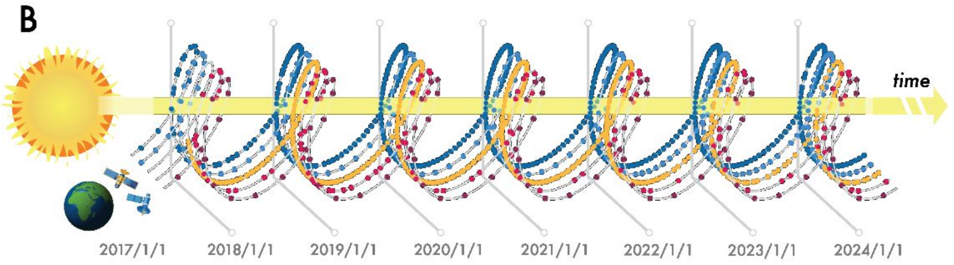
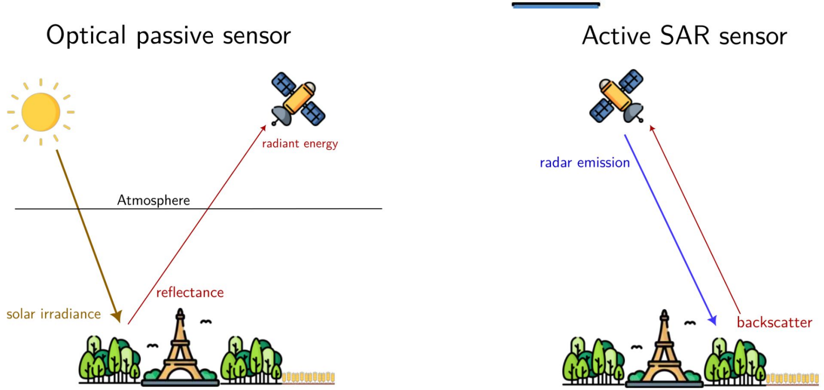
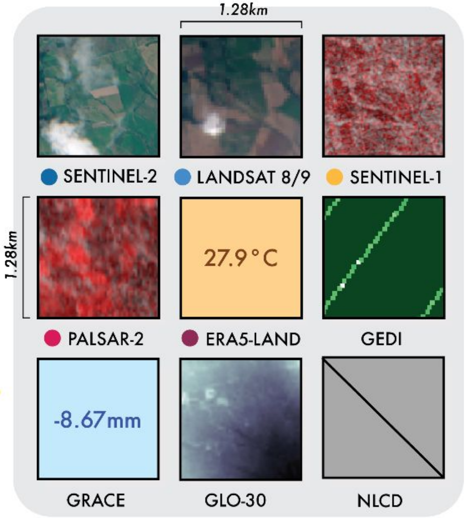
Introduction

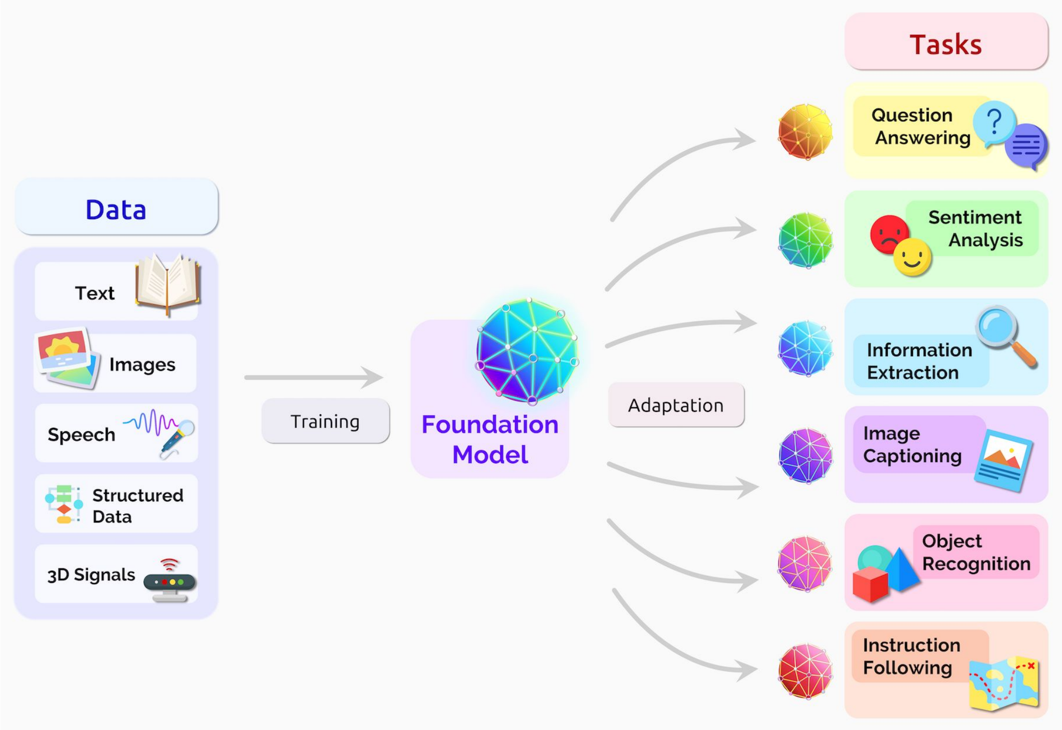
Remote sensing time series and foundation models



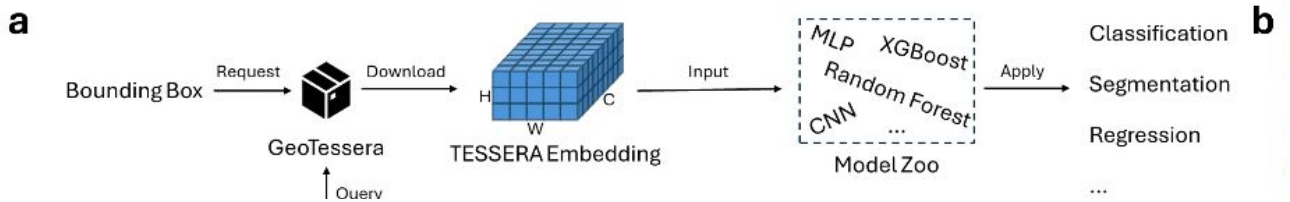




Remote sensing time series and foundation models



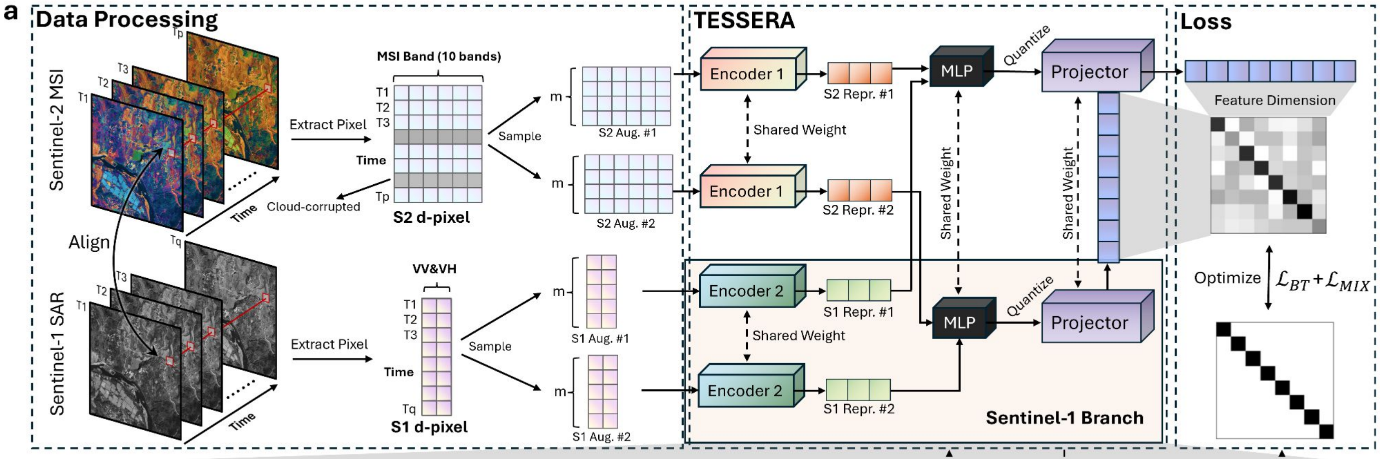
Foundation models in remote sensing data



Zhengpeng Feng et al. TESSERA: Temporal Embeddings of Surface Spectra for Earth Representa-tion and Analysis. 2025. arXiv: 2506.20380 [cs.LG]. URL: https://arxiv.org/abs/2506.20380

Methods

TESSERA: Temporal Embeddings of Surface Spectra for Earth Representation and Analysis

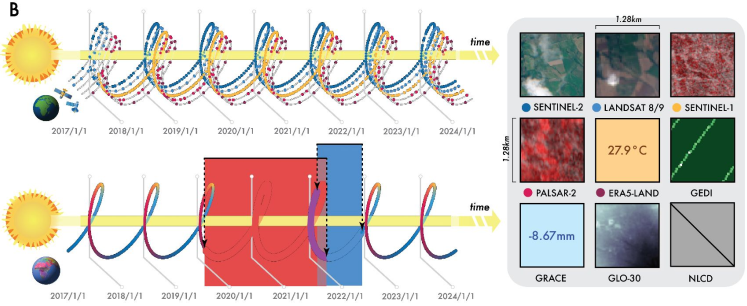


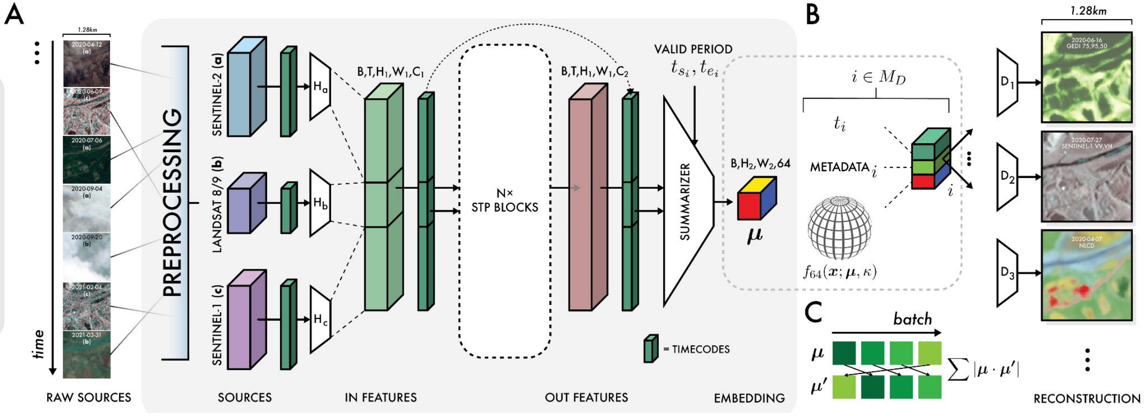
Core Concept: a pixel-wise Remote Sensing Foundation Model (RSFM) for multi-modal (Sentinel-1 & 2) Earth Observation.
Key Innovation: uses "temporal sampling invariance" (via Barlow Twins contrastive learning) to handle irregular, cloud-obstructed satellite time series naturally.

$$\mathcal{L}_{BT} = \sum_i (1 - C_{ii})^2 + \lambda_{BT} \sum_i \sum_{j \neq i} C_{ij}^2$$

Zhengpeng Feng et al. TESSERA: Temporal Embeddings of Surface Spectra for Earth Representa-tion and Analysis. 2025. arXiv: 2506.20380 [cs.LG]. URL: https://arxiv.org/abs/2506.20380

Alpha Earth





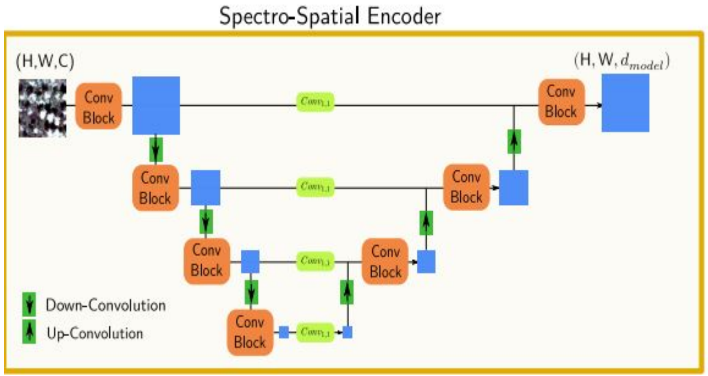
Multi-Modality: Fuses optical (Sentinel-2, Landsat), radar (Sentinel-1, PALSAR-2), LiDAR (GEDI), environmental data (climate, gravity, elevation), and geocoded text (Wikipedia, GBIF).

Learning Strategy: Generative/reconstruction framework utilizing an implicit decoder and a spatially dense information time-bottleneck, co-trained with a text-contrastive objective.

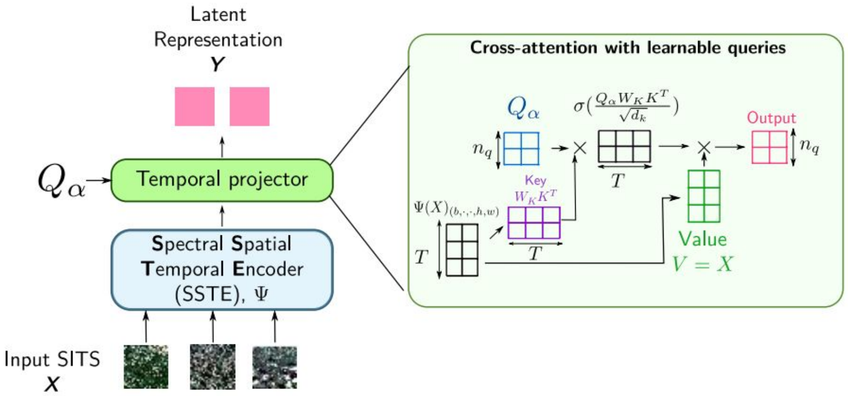
Christopher F. Brown et al. AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data. 2025. arXiv: 2507.22291 [cs.CV]. URL: https://arxiv.org/abs/2507.22291

ALISE

U-BARN Architecture



ALISE Architecture



ALISE : novel encoder designed to transform irregular and unaligned satellite image time series into aligned, fixed-size representations while preserving high spatial resolution

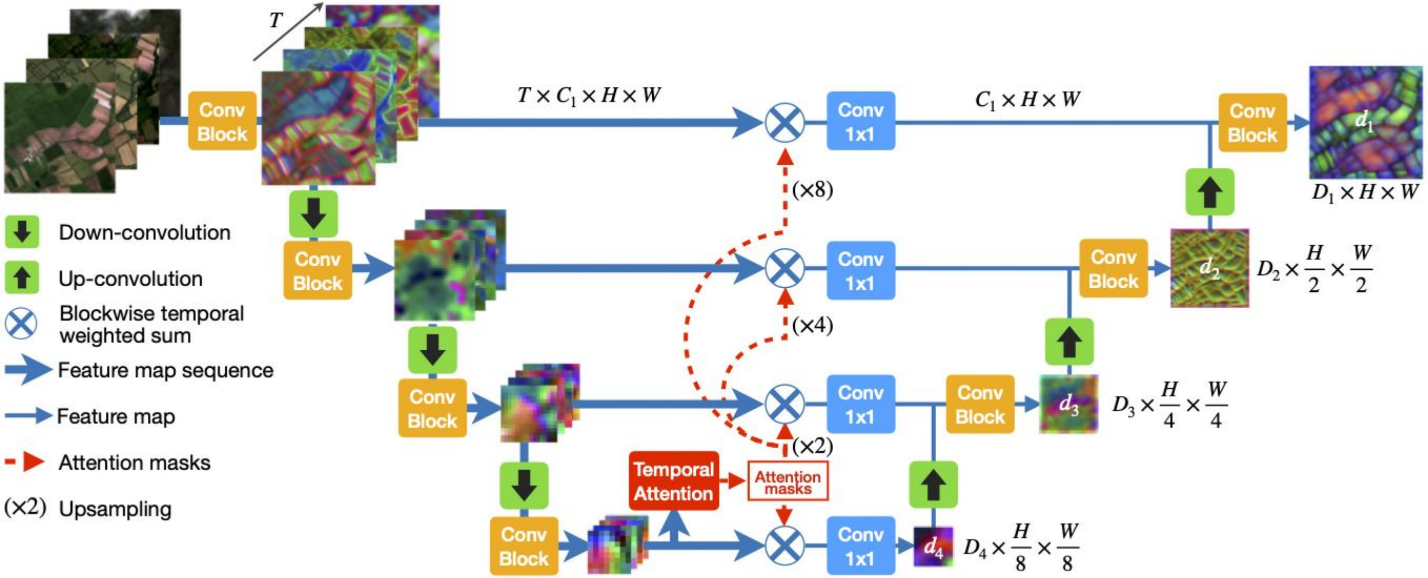
I. Dumeur, S. Valero, and J. Inglada, "Self-Supervised SpatioTemporal Representation Learning of Satellite Image Time Series

Iris Dumeur et al. Paving the way toward foundation models for irregular and unaligned Satellite Image Time Series. 2024

FM Comparison table

	Tessera	Alpha Earth	Alise
Input	SAR (S1) and 10-band optical image (S2)	SAR (S1), 10-band optical image (S2), Thermal (Landsat-8/9)	10-band optical image (S2)
SSL	Contrastive	Hybrid generative	Hybrid generative
Spatial feature extraction	❌	✅	✅
Embedding size	128	64	640

U-TAE



U-TAE: A shared convolutional encoder processes images in parallel, using temporal attention masks to collapse time into single feature maps before a convolutional decoder

Garnot, V. S. F. et Landrieu, L., ***Panoptic Segmentation of Satellite Image Time Series with Convolutional Temporal Attention Networks***

Benchmarking and results

Dataset : PASTIS

-Primarily Sentinel 2 multispectral data

- 4000 km^2 covered in 4 French regions

-More than 120 000 parcels annotated

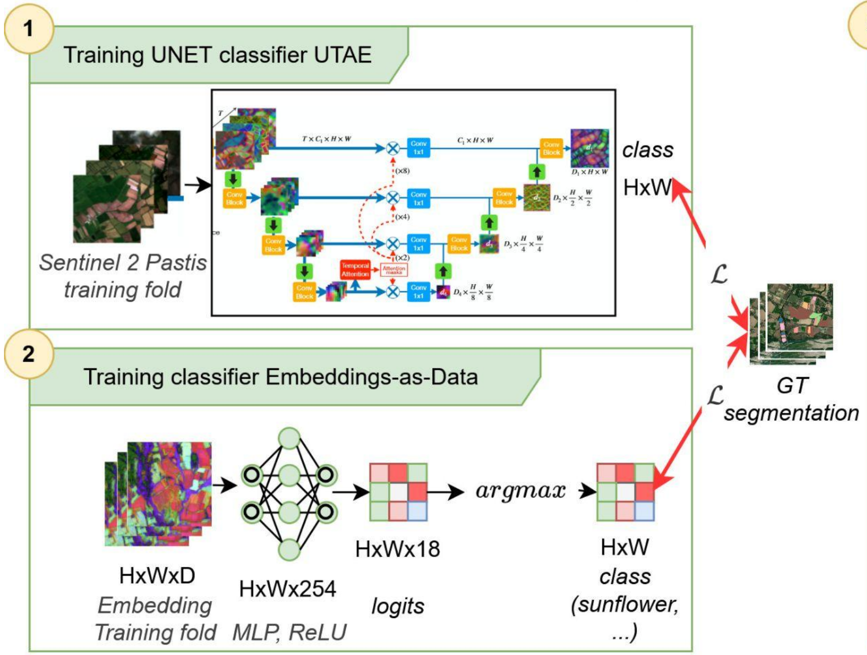
-18 crops classes + Background

-between 38 and 61 images per year

Satellite time series Images

Task and experiment setup

$$\mathcal{L} = \frac{1}{|\Omega'|} \sum_{(b,h,w)\in\Omega'} \ell_{CE}(z_{b,h,w}, y_{b,h,w})$$



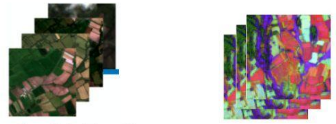
3

Testing

$$\text{Accuracy} = \frac{1}{|\Omega'|} \sum_{(b,h,w)\in\Omega'} \mathbf{1}\left(\hat{y}_{b,h,w} = y_{b,h,w}\right)$$

$$\text{F1}_{\text{macro}} = \frac{1}{K} \sum_{k=1}^{K} \frac{2\,TP_k}{2\,TP_k + FP_k + FN_k}$$

$$\text{mIoU} = \frac{1}{K} \sum_{k=1}^{K} \frac{TP_k}{TP_k + FP_k + FN_k}$$

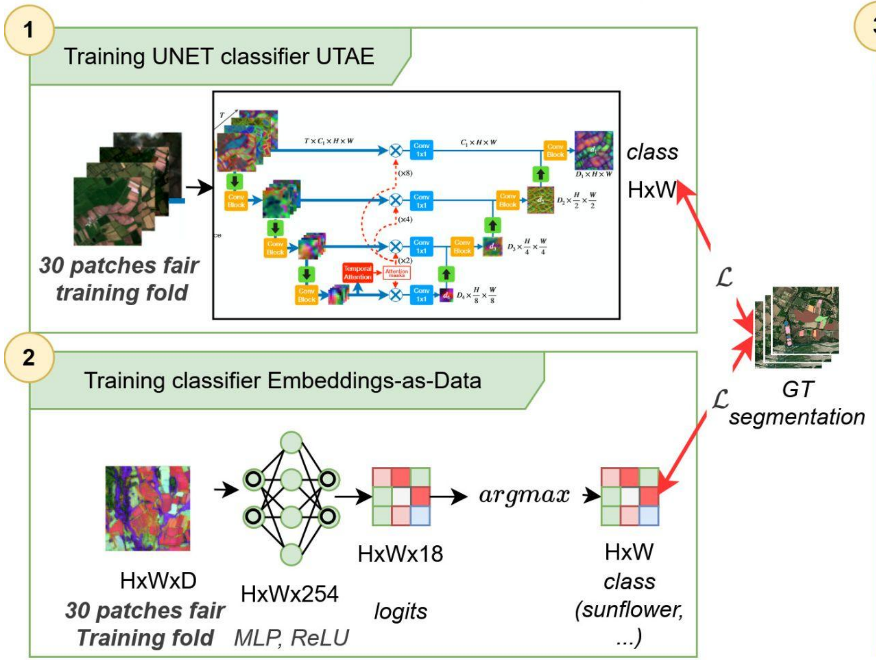


Weights & Biases

MesoUPSaclay (4 GPUs V100, 40 cores, 200 epochs, 20 minutes training)

Second experiment setup

$$\mathcal{L} = \frac{1}{|\Omega'|} \sum_{(b,h,w)\in\Omega'} \ell_{CE}(z_{b,h,w}, y_{b,h,w})$$



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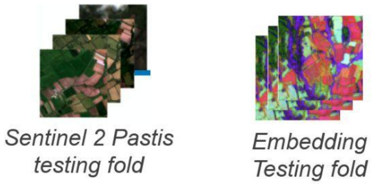


Weights & Biases MesoUPSaclay (4 GPUs V100, 40 cores, 200 epochs, 20 minutes training)

Results

Table 1 Test metrics

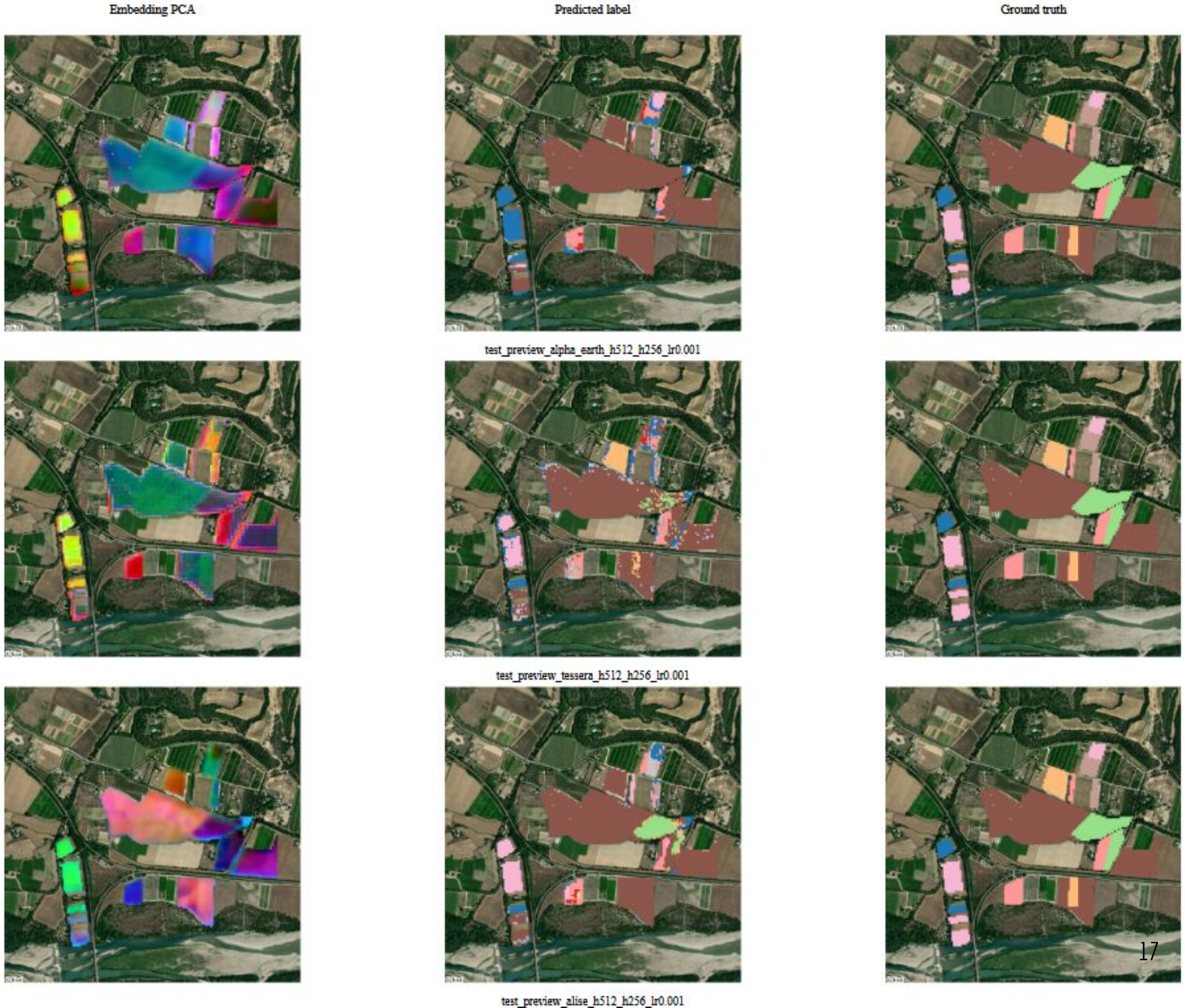
Model	mIoU	F1 Macro	Overall accuracy
AlphaEarth	0.5287	0.6526	0.8417
Tessera	0.5607	0.7018	0.8257
ALISE	0.6009	**0.7235**	0.8651
U-TAE	**0.7219**	-	**0.9170**

Table 2 Test metrics with scarce data (30 patches per fold)

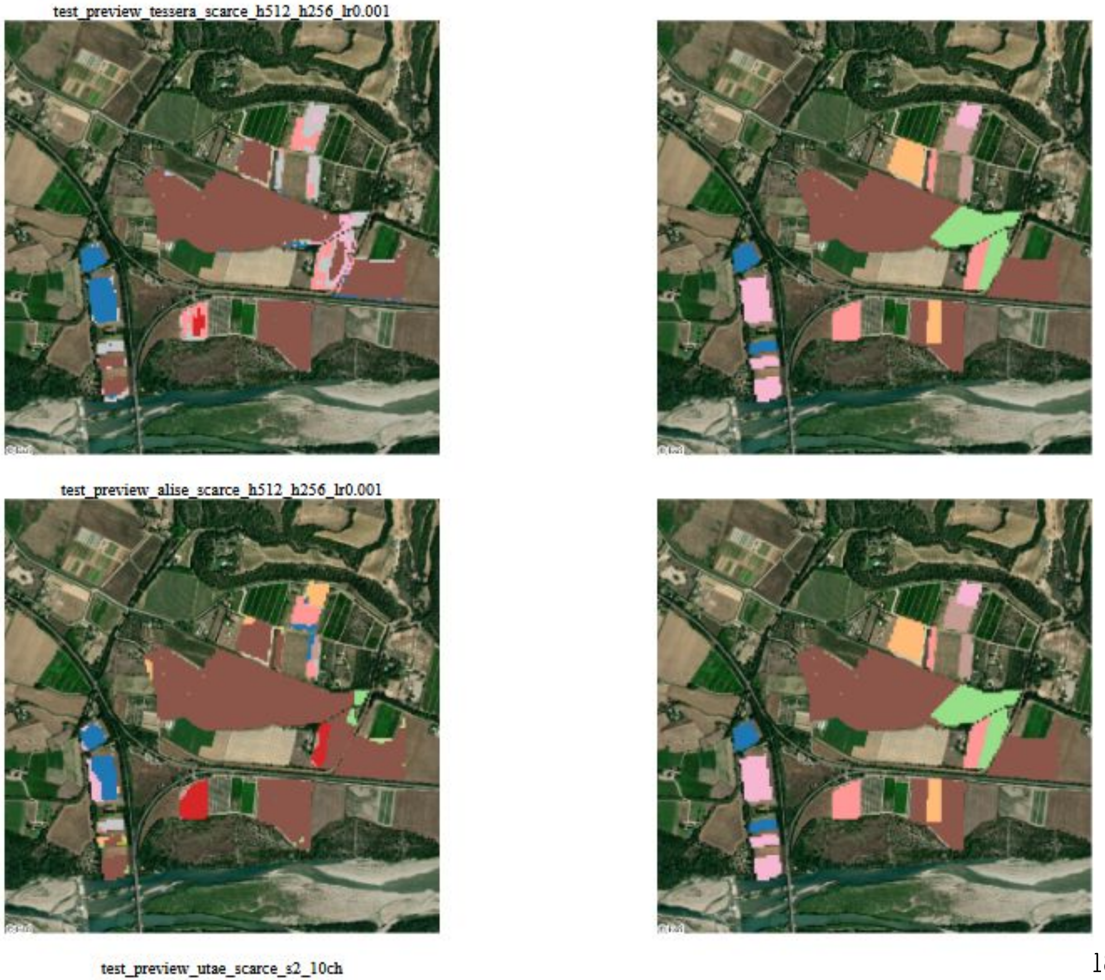
Model	mIoU	F1 Macro	Overall accuracy
AlphaEarth	0.2867	0.3755	0.7267
Tessera	0.4387	**0.5765**	0.7621
ALISE	**0.4482**	0.5729	**0.7898**
U-TAE	0.3869	0.5115	0.7043

First experiment

PC1 (Red)
PC2 (Green)
PC3 (Blue)
Meadow
Winter barley
Spring barley
Sunflower
Grapevine
Winter triticale
Winter durum wheat
Fruits, vegetables, flowers
Leguminous fodder
Orchard



Second experiment



Next steps and conclusion

- What-is-this-crop.net
- Try with...
  - Other tasks
  - Other geographic zones
  - Other FM models
- Multimodality